Conversion of Camphene to Camphor: Day 3

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Introduction

In this experiment, borneol was converted to camphor through reflux distillation. Borneol was combined with glacial acetic acid and then dissolved in acetone. Reflux distillation was then performed and bleach was added slowly. Camphor was extracted with methylene chloride, or DCM, then was neutralized with saturated sodium bicarbonate. The solution was washed with 5% sodium bisulfite and water. Anhydrous sodium sulfate was used to dry the organic solution and the DCM was evaporated over a water bath until the product was solid camphor.

Materials and Methods

In a long-neck 10 mL round-bottom flask, 0.3 g of borneol product was added to 0.3 mL of glacial acetic acid. The racemic mixture was then dissolved in 1 mL of acetone. A boiling chip was added. A reflux with distillation column was assembled and placed in a water bath with the temperature between 40-50 degrees celsius. Once the reagents were dissolved, 6 mL of aqueous sodium hypochlorite solution, or bleach, was added to the mixture. The addition was done through the top opening of the condenser and was paced dropwise over a 10 minute period. After each addition of the bleach, the mixture was swirled to ensure that all chemicals were mixed thoroughly. The reaction was left to continue for 15 minutes after the last bleach addition. The mixture was then left to cool to room temperature. Camphor was extracted with two portions, 2 mL each, of methylene chloride into a centrifuge tube. The combined methylene chloride layers were then neutralized with 2 mL of saturated sodium bicarbonate. The organic solution was washed sequentially with 2 mL of 5% sodium bisulfite and 2 mL of water. The solution was then dried over anhydrous sodium sulfate until freely rolling crystals were observed. DCM was evaporated over a water bath with the temperature not exceeding 50 degrees celsius. The collected crystals were weighed and percent yield was determined.

Results

Weight of Watch Glass	28.503 g
Weight of Watch Glass & Product	28.749 g
Weight of Product	0.246 g

Table 1. Weight of watch glass, watch glass & product, and product.

Calculations

Actual Yield = **0.246 g**

Theoretical Yield

D = m / v → 1.01 g/mL = m / 0.741 mL → m = 0.748 g

m / mm borneol = mols → 0.748 g / 154.25 g/mol = 0.00485 mols

mols (mm camphor) = grams → 0.00485 mols (152.23 g/mol) = **0.738 g**

% Yield = 0.246 g / 0.738 g = 0.333 x 100 = **33.3 %**

Discussion

In this experiment, 0.246 g of camphor was produced from borneol through reflux distillation. The percent yield calculated suggests that a large amount of product was lost during reflux. Further analysis could be taken to determine the composition of the product and why material was lost. Possible sources of error include pipetting the wrong layer, not refluxing long enough, and overheating the mixture during reflux. Of the sources of error mentioned, the one to have most likely occurred would be pipetting the wrong layer. Pipetting the right layer is determined by density. The product with the larger density sits on the bottom, so the product's

density should be known in order to extract the correct layer. One way to resolve this error is to write down the densities of the reactant and product before starting the experiment.

References

Keller L, Joseph K, and Miguel S. 2022 Fall. CHEM 2081 Organic Chemistry 1 Lab Manual [lab manual]. Auburn (Al): Auburn University.